

**Table 1. False-premise detection across models and datasets**

| Method                     | Qwen2.5 7B                                                                                    |                   |       |       | GPT-6 Luna  |       |       |       | Gemma 4 12B       |                   |       |       | Qwen3.8 27B OFF   |                   |       |       | Qwen3.8 27B ON    |                   |       |       |
|----------------------------|-----------------------------------------------------------------------------------------------|-------------------|-------|-------|-------------|-------|-------|-------|-------------------|-------------------|-------|-------|-------------------|-------------------|-------|-------|-------------------|-------------------|-------|-------|
|                            | Cancer-Myth                                                                                   |                   | CREPE |       | Cancer-Myth |       | CREPE |       | Cancer-Myth       |                   | CREPE |       | Cancer-Myth       |                   | CREPE |       | Cancer-Myth       |                   | CREPE |       |
|                            | TPR ↑                                                                                         | FPR ↓             | TPR ↑ | FPR ↓ | TPR ↑       | FPR ↓ | TPR ↑ | FPR ↓ | TPR ↑             | FPR ↓             | TPR ↑ | FPR ↓ | TPR ↑             | FPR ↓             | TPR ↑ | FPR ↓ | TPR ↑             | FPR ↓             | TPR ↑ | FPR ↓ |
| Direct gate                | 7.0 <sup>H</sup>                                                                              | 10.7 <sup>H</sup> | —     | —     | —           | —     | —     | —     | 40.5 <sup>H</sup> | 16.1 <sup>H</sup> | —     | —     | 26.1 <sup>H</sup> | 16.1 <sup>H</sup> | —     | —     | 44.1 <sup>H</sup> | 21.5 <sup>H</sup> | —     | —     |
| Premise-review CoT gate    | 29.5 <sup>H</sup>                                                                             | 16.1 <sup>H</sup> | —     | —     | —           | —     | —     | —     | 51.3 <sup>H</sup> | 32.9 <sup>H</sup> | —     | —     | 57.5 <sup>H</sup> | 35.6 <sup>H</sup> | —     | —     | 61.4 <sup>H</sup> | 38.3 <sup>H</sup> | —     | —     |
| Probe gate (hidden states) | 75.0 <sup>H</sup>                                                                             | 47.7 <sup>H</sup> | —     | —     | n/a         | n/a   | n/a   | n/a   | —                 | —                 | —     | —     | —                 | —                 | —     | —     | —                 | —                 | —     | —     |
| TF-IDF text                | Shared across models — Cancer-Myth: TPR ↑ 86.1% (H), FPR ↓ 51.0% (H); CREPE: TPR ↑ —, FPR ↓ — |                   |       |       |             |       |       |       |                   |                   |       |       |                   |                   |       |       |                   |                   |       |       |
| GEPA                       | —                                                                                             | —                 | —     | —     | —           | —     | —     | —     | —                 | —                 | —     | —     | —                 | —                 | —     | —     | —                 | —                 | —     | —     |
| Ours                       | —                                                                                             | —                 | —     | —     | —           | —     | —     | —     | —                 | —                 | —     | —     | —                 | —                 | —     | —     | —                 | —                 | —     | —     |

Stored results; —: not measured/comparable; n/a: outside the defined implementation. Percentages, one run.  
H: historical full-corpus Cancer-Myth; valid denominators and protocol differences in ledger.  
TPR ↑: FPQ detection; FPR ↓: false positives on normal questions.  
Probes require hidden states; no native Luna probe. TF-IDF is model-independent.  
Default crossfit probe/TF-IDF decisions; these are not matched-FPR thresholds.  
Ours denotes the proposed method; its implementation is not fixed.